

Sample Program: Matrix Multiplication in Python

Input Format (Matrix A and Matrix B):

A,i,j,value
B,j,k,value

mapper.py

```
#!/usr/bin/env python3
import sys

for line in sys.stdin:
    tokens = line.strip().split(',')
    if tokens[0] == 'A':
        for k in range(0, 2): # assuming output matrix has 2 columns
            print(f"{tokens[1]},{k}\tA,{tokens[2]},{tokens[3]}")
    elif tokens[0] == 'B':

        for i in range(0, 2): # assuming output matrix has 2 rows
            print(f"{i},{tokens[2]}\tB,{tokens[1]},{tokens[3]}")
```

reducer.py

```
#!/usr/bin/env python3
import sys
from collections import defaultdict

current_key = None
A_vals = defaultdict(float)
B_vals = defaultdict(float)

for line in sys.stdin:
    key, value = line.strip().split('\t')
    matrix, index, val = value.split(',')
    if current_key and current_key != key:
        result = sum(A_vals[k] * B_vals[k] for k in A_vals if k in B_vals)
        print(f"{current_key}\t{result}")
        A_vals.clear()
        B_vals.clear()
    current_key = key
    if matrix == 'A':
        A_vals[index] = float(val)
    else:
        B_vals[index] = float(val)

if current_key:
    result = sum(A_vals[k] * B_vals[k] for k in A_vals if k in B_vals)
    print(f"{current_key}\t{result}")
```

Input Files in HDFS:

Matrix A (a.csv)

```
A,0,0,1
A,0,1,2
A,1,0,3
A,1,1,4
```

Matrix B (b.csv)

```
B,0,0,5
B,1,0,6
B,0,1,7
B,1,1,8
```

Execution Commands:

```
# Upload files to HDFS
hadoop fs -put a.csv /input/a.csv
```

```
hadoop fs -put b.csv /input/b.csv
```

```
# Run MapReduce Job
hadoop jar /usr/lib/hadoop-mapreduce/hadoop-streaming.jar \
-files mapper.py, reducer.py \
-mapper mapper.py \
-reducer reducer.py \
-input /input \
-output /output_matrix
```

```
# View result
hadoop fs -cat /output_matrix/part-00000
```

Expected Output:

```
0,0\t17.0
0,1\t23.0
1,0\t39.0
1,1\t53.0
```